

A Review of Ceramic Sintering and Suggestions on Reducing Sintering Temperatures

X. KUANG, G. CAROTENUTO AND L. NICOLAIS*

Department of Materials Engineering and Production, University of Naples, Piazzale Tecchio—80125, Napoli, Italy

Abstract. Theories and applications associated with sintering of ceramics and reducing of sintering temperature are reviewed. The whole sintering process is divided into three sub-processes as powder preparation, compaction and sintering, and each sub-process is discussed in terms of its possible contribution to the reducing of sintering temperature. New approach for practical optimization of sintering process is investigated. The application of above in the Low Temperature Co-fired Ceramics (LTCC) is discussed. Meanwhile, many successful applications in reducing the sintering temperatures are presented.

Keywords: ceramics, controlled double-jet, hydrothermal processes, nano-size particles, sintering, surface activation

1. Introduction

Sintering was considered to be the symbol of human civilization, which started with firing of pottery and earthenware in ancient times [1]. Nowadays, sintering is widely used to make useful parts from either metal or ceramic powders [2]. The powder particles bond to each other during firing and matter transport occurs. The micro-pores are eliminated and the piece becomes denser and stronger. A great variety of technologically important products are made by sintering. Cemented tungsten carbide cutting tools and the silicate-bonded alumina used extensively in electronics use a liquid phase during sintering. More examples of solid-state sintering are ferrite magnets, translucent alumina envelopes for high-pressure sodium vapor lamps, uranium dioxide fuel for nuclear reactor and hard silicon carbide wear parts.

The common feature of all sintering processes involves the using of high temperature, though some nanophase particles are reportedly sintered to close to full density at room temperature or a few hundred degrees [3, 4]. Generally, the normal sintering temperatures of ceramics range from 1000°C to more than 2000°C [5]. The sintering temperature that has experimentally established was because materials can be sintered in a short time of one or two hours. The disadvantages of using a high sintering temperature include high production cost, high maintenance fee, difficult quality control and high equipment cost. Recently, more and more concern has been paid to the high energy consumption, and the excessive growth of grain size during high temperature sintering process.

*Currently with American Electronic Materials (AEM), Inc., San Diego, California.

Lower temperature sintering is of both profound theoretical and practical meanings. How to improve the sinterability of ceramics while maintaining quality is an interesting topic that is attracting much attention in both academia and industry.

There are many factors governing the sintering temperature of ceramics. As considered in theory, one can divide these parameters into two classes [1]:

- a) *Intrinsic*. These specify the intrinsic properties of the materials being sintered, such as surface free energy, diffusion coefficient, vapor pressure, viscosity etc. These parameters change with the chemical composition, ambient atmosphere or temperature changes;
- b) *Extrinsic*. These depend on the geometrical or topological details of the whole sintering system. These include average particle size, size distribution, particle morphology and green body density.

However, the real sintering system is far more complicated, and very often the sintering problems are discussed in terms of the following three categories [6]:

- a) *Powder*. Particle size, size distribution, particle morphology, inter-particle porosity, agglomeration, particle chemical homogeneity, absorbed gases, impurities and surface chemistry;
- b) *Pressed compact*. Packing density, pore distribution, pore size distribution, effects from granulation, crystallographic texture;
- c) *Sintering*. Temperature, temperature gradient, gas atmosphere, second phase, realized activated sintering and pressure.

Among them, particle size, size distribution, morphology, surface chemical properties and green body density are frequently considered. What is most wanted is to achieve the highest final product quality and the fastest development and production at the lowest total cost. As far as lower sintering temperature is concerned, it is reasonable to discuss all of the categories.

2. Recent advance in sintering theory

Based on the careful analysis of the sintering process, Hausner [7] said “sintering is the bonding of particles in a mass of powders by molecular or atomic attraction in the solid state, by application of heat, causing strengthening of the powder mass and possible resulting in densification and recrystallization by transport of materials.” From a different point of view, Kuczynski [8] defined sintering as a diffusional creep under action of capillary forces, which is more meaningful in processes like “activated sintering.”

The first theory of sintering was rendered by Frenkel, Pines, Ivensen and Kuczynski in the middle of this century [1]. Their research work and scientific activities have been continued and further developed by Geguzin, Coble, Johnson, Skorohod, Shatt and many others [9, 10]. All their investigations were based on atomistic processes in a solid state, in which principles still present the basis of the physics of sintering. According to their theory, the sintering process can be broadly placed into three stages:

- a) *Early stage*. In this stage, crystallites do not lose their individuality, nor merge or recrystallize, and no appreciable shrinkage of compacted powder is observable. This

stage occurs when the substance in question attains 1/4 of its melting point in Kelvin degrees;

- b) *Intermediate stage.* This stage consists of a comparatively free wandering of lattice elements, including lattice holes on the surface, and is characterized by a formation of mutually adhering particles which lead to the densification and grain growth. This takes place at a temperature 1/4–3/4 of the °K melting point. Here, the first sign of shrinkage appears. At this stage, different crystallographic planes will touch each other. Since such planes possess different surface energies, the plane of a higher energy level will feed the plane of a lower level. Therefore, the latter grow at the expense of the first. Normally, the surface energy of very small particles is higher than that of coarser grains, which results in the disappearance of fine particles and growth of coarse grains;
- c) *Final stage.* At this stage, densification process will finish and micro-pores will close.

All the above stages can be perfectly described in terms of porosity and pore characteristics.

Sintering mechanisms have been investigated by various workers under two main categories, plastic deformation [11–14] and diffusion [15, 16]. The plastic deformation occurs mostly in the early stage. At the intermediate and final stage, the diffusion at an atomistic level controls the sintering process.

Chaklader reported that the diffusion on grain boundaries because of volume diffusion became the controlling sintering mechanism when the particle size decreased [17]. Starting with this fact, Chu [18] obtained a common equation to characterize the condensing density during the process:

$$\varepsilon_d = \frac{HD\phi^{\frac{(n+1)}{2}}(S+P)}{x^n kT} \quad (1)$$

where: ε_d —the linear densification strain rate, H —a constant, D —the diffusion coefficient, S —the sintering stress, P —an additional hydrostatic applied stress ($P = 0$ for free sintering), ϕ —the stress intensification factor, $n = 3$ for grain boundary diffusion, $n = 2$ for volume diffusion, x —the center-to-center pore spacing, k —Boltzman's constant and T —the absolute temperature. From the Eq. (1), the sintering temperature can be preliminarily predicted.

With the development of fast speed computers, the numerical simulation of the sintering process becomes more and more popular. It provides ceramic scientists and engineers a chance to visualize the process and evolution of the micro-structure. Srolovitz developed a Monte Carlo model for simulating the final stage sintering [19, 20]. This model incorporated realistic micro-structure features (grains and pores), variable surface diffusion, grain boundary mobility. A study of a period array of pores has shown that the simulation procedure accurately reproduces the theoretically predicted sintering kinetics under the restricted set of assumptions. The porosity decreases monotonously with sintering time and diffusion along the grain boundary.

The Monte Carlo simulation was also successfully applied to the initial sintering stage of randomly packed particles by Madhavrao and Rajagopalan [21]. In their paper, the possibility of developing local stress and stress propagation was introduced in terms of a probability formulation, which can be related to the changes in local geometry, material

properties and any relevant thermodynamic considerations. The proposed model can offer a convenient way to study the effects of particle rearrangements, packing characteristics and possibility of crack development during sintering process.

The same simulation procedure has been used to investigate the effect of initial pore size on the micro-structure evolution and kinetics of final sintering stage [20, 22]. Pores were found to effectively stabilize the grain boundaries from the beginning of final stage sintering at a porosity $\Phi = 0.09$ until $\Phi \leq 0.03$. For $\Phi = 0.13$, the remaining pores do not effectively retard grain boundary migration and normal grain growth occurs.

The contribution of atomic theory of sintering is so outstanding which can explain most of the sintering behavior. Further comprehension of the essence of the sintering process may start from the electronic point of view that scientists are still far away from an uniform thorough solution. The pioneering work by Samsonov [23], German [24] and Uphadhyaya [25] explored a way to the solution.

From quantum mechanics point of view, sintering should be treated as a process of approaching equilibrium state of electronic-nuclei-system that contains irregularities in ideal and periodical distribution of atoms. Quantum theory on chemical reactions in condensed state can describe deviation of a system from equilibrium state and possible ways of their return to that state [10]. Using the Schrödinger equation,

$$H(r, R)\Psi(r, R) = \varepsilon(R)\Psi(r, R) \tag{2}$$

The electronic subsystem can be solved for arbitrary distribution of nuclei within a configuration space (R) of coordinates of all particles entering the system. In above expression, $H(r, R)$ is system of Hamiltonian operators that depend on the sum of nuclei coordinates (R). Electron coordinates (r), $\varepsilon(R)$ is energy of the system that matches the given configuration of cores, and $\Psi(r, R)$ is the wave function of the system.

The dependence of ceramic properties on sintering process can be described as a sum of “electronic structure—atomic defects—crystal nano-structure—micro-structure—macro-structure—properties”. The structure hierarchy is presented in figure 1.

A detailed (final) shape of a hyper-surface of the potential energy $\varepsilon(R)$ is determined by chemical nature of an element. Therefore, so far the application of this theory is limited to a description of pure elements of the Periodic Table System. In practice, a general quantum-mechanical determination of the $\Psi(r, R)$ and $\varepsilon(R)$ is extremely difficult to achieve. Most of these values can be quantitatively presented using an essential hypothesis based on quantum chemistry and solid-state physics. Nevertheless, there is still a long way to go for quantum sintering theory before it becomes a practical tool for ceramic scientists and engineers.

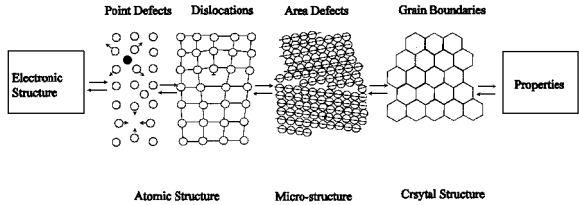


Figure 1. The structure hierarchy of sintering.

From the available sintering theory, one can probably get a rough idea about how to reduce the sintering temperature while maintain the quality. However, in real systems the geometric and chemical assumptions described in the above theories do not exactly apply. As stated in the introduction, the dependence on temperature of sintering process can be divided in terms of three independent processes, namely powder, compaction and sintering. In the real sintering production of ceramics, one often has to consider all the influential factors, though some factors may be more important than the other. Therefore, the whole sintering process is investigated in terms of powder, pressed compact and sintering to discuss possible contribution to reduce the sintering temperature.

3. Powder I: starting powders manufactured by conventional ceramic process

Most of the raw materials used in the ceramic industry today are obtained from conventional ceramic methods [26]. For example, Ni-Zn ferrites are produced by mixing stoichiometrical amounts of nickel, zinc and iron oxides or carbonates oxide, and the mixture is fired at 1000°C to finish the solid-state reaction [27]. Then the resultant ferrite solid will go through all the conventional mechanical size-reduction processes and be spray-dried to get final powders. The average size obtained is usually at the order of microns depending on the used process, and the sintering temperature is around 1200°C. To get a desired sintered body at a lower sintering temperature, there is still some work that can be done.

First, it is important to get as small particle size, agglomeration free particles and uniform size distribution as possible at reasonable process cost. As it is well known, the general driving force at the ceramic sintering is the surface free energy of particles [28]. The smaller the particle size, the higher the surface free energy, the easier of surface atom diffusion between neighboring particles, and the lower sintering temperature. With the narrow size distribution, the higher green body density can be achieved, and correspondingly the total contact area and the number of particle lattice imperfections are increased; though in certain cases, e.g., the ideal compaction system of binary particle size, the highest packing density can only be reached by certain combination of those two component particles. Higher contact surface area will enhance the surface diffusion. In fact, lattice imperfections are considered the second driving force in sintering [28]. It should be borne in mind that the lattice imperfections are the main cause to break solids into particles under external force. Therefore, the process of comminution process exposes the imperfections, previously hidden inside the large grain, bringing them to the surface of small particles. Accordingly, such freshly produced small particles are double charged with free energy enabling them to be integrated at lower sintering temperature.

Adding certain chemicals, which are often surface active, to the mill feed in small quantities (0.1 wt% of the feed solids) can improve the grindability of solids. Those chemicals are called “grinding aids” [29–33]. The mechanisms involved in grinding aids action may include 1) the reduction of breakage energy, 2) embrittlement to reduce plastic deformation, 3) the flocculation/dispersion of fines, 4) the prevention of reagglomeration, and 5) the control of slurry rheology.

As far as the reduced surface energy by grinding aids is concerned, the Griffith theory for the energy balance in crack extension has been successfully applied. Based on the theory, a

Table 1. Surface energy lowering and fracture strength of silica rods in various saturated vapors.

Saturated vapor	Surface free energy lowering, mJ/m ²	Fracture strength, MPa
Vacuum	0	91.1 ± 4.8
Dry benzene	52	93.9 ± 6.9
Butyl alcohol	85	82.2 ± 4.1
Propyl alcohol	110	66.3 ± 0.7
Water	244	55.2 ± 3.9
Water*	244	45.6 ± 4.8

*This series of breakage tests was carried out after 100 minutes of exposure to water vapor, whereas all of the other results were for 10 minutes of exposure.

crack will propagate if the loss in elastic strain energy is greater than the increase in surface energy accompanying the formation of new surfaces. For a crack of length $2a$ inside a specimen, the fracture strength, σ_f , is given by [34]:

$$\sigma_f = \frac{2Y\gamma^{1/2}}{\pi a} \tag{3}$$

where Y is the module of elasticity and γ is the surface free energy of the solid material. The equation tells us that the crack energy will decrease with the reduction of surface energy.

Hammond and Ravitz [35] measured the tensile breaking load of annealed fused silica rods in various saturated vapors, and found that the breaking load decreased in accordance with the reduction in surface tension caused by the adsorption of vapor. The Gibbs adsorption equation was used to calculate the lowering of the surface tension of the solid. Table 1 summarizes their results.

The results given in Table 1 show a time effect. Since the diffusion of molecules to the crack tip and possible chemical reaction requires a finite time, the environment can affect crack propagation only if the crack velocity is sufficiently low (as in controlled cleavage experiments).

For more information about grinding aids, their theory and ultrafine particle grinding, please refer to references [36–45].

As it is well known that, the main driving force in sintering is the high surface free energy of the fine particles [2, 28]. The surface free energy increases quickly with the decrease of particle size. Correspondingly sintering temperature of ultrafine particles is lower than that of larger particles, if both of the particles have the same degree of agglomeration. Based on this principle, ultrafine grinding in the submicron range has currently attracted attention in connection with the development of new ceramics and electronic materials. Many investigators have reported experimental data of wet process milling using grinding media. As mills using grinding media, conventional ball mills, vibrational mills, planetary mills and stirred mills are typical machineries; special interests have been focused on ultrafine grinding using grinding balls smaller than 1 mm in diameter. Moreover, a specific ball size distribution may lead to a remarkable improvement in the rate of milling [46]. Besides used to improve grinding rate, grinding aids are also used to modify of the finished surface of the ground particles.

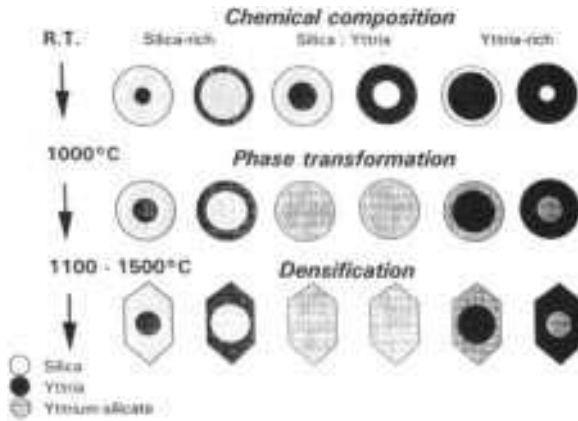


Figure 2. Evolution of the grain boundary in the sintering systems of yttria-coated-silica and silica-coated-yttria.

Second, sintering additives (0.005–0.1 wt%) can improve sinterability of ceramics [47–50]. This effect is mainly due to the combination of additives and core particles, which will form a low melting point compound or something like glass. The sintering additives are also called sintering aids or fluxing agents. Practically, 0.005 wt% BiO_3 [51, 52] was applied in the sintering of ferrites, and improved densification and homogenization of resultant sintered body and avoiding oxygen losses were observed. In other sources, CaO and SiO_2 were also used as sintering additives [48, 49]. Moreover, the additives are normally distributed around grain boundaries, and will inhibit the growth of grain size. The advantages of using sintering additives include the lower sintering temperature (less energy consumed and easier quality control), higher resistivity, fine grain size (which is very useful for the high frequency applications) and improved permeability. However, there is a certain limit of using sintering aids, e.g., the permeability will decrease after the content of SiO_2 or CaO is more than 0.2 wt%. The idea of using sintering aids to improve sinterability was adopted by Gieche to sinter Y_2O_3 with a thin layer coating of SiO_2 to full density at only 1250°C [50, 53]. Meanwhile, Gieche also offered an explanation for the evolution of the grain boundary as present in figure 2. Table 2 shows some commonly used sintering aids and grain-growth inhibitors.

As indicated above, there are at least two methods to apply sintering additives into pre-sintering powders. One is to mix the additives directly, and the other is to get a thin layer coating of the desired additives by chemical methods. The coating process can be done in a lot of different ways, and some examples are shown in figure 3.

Finally, particle surface modification can improve sinterability. For example, Plasma Activated Sintering (PAS) is performed in an air atmosphere using a processing schedule consisting of a powder activation step and a sintering step. It was reported that aluminum oxide can be sintered to full density at 1150°C for 10 minutes [54]. In the reported experiments, activation step involved applying dc electric discharge pulses (voltage 25 V and current 750 A) that stayed on for 30 ms and off for 30 ms. These 30 ms on/off electric discharges pulses were applied for a total duration varying from 60–90 seconds, and then the desired plasma was generated between powder particles by the high current pulse cycles. As a result of the activation process, the surface of powders was activated, in another words, the surface possessed higher free energy than it would have without activation.

Table 2. Sintering aids and grain-growth modifiers.

Materials	Sintering aids	Grain-growth inhibitors
Al ₂ O ₃	LiF	Mg, Zn, Ni, W, BN, ZrB ₂
MgO	LiF, NaF	MgFe, Fe, Cr, Mo, Ni, BN
BeO	Li ₂ O	Graphite
Si ₃ N ₄	MgO, Y ₂ O ₃ , BeSiN ₂	—
SiC	B, Al, Al ₂ O ₃	—
TiC, WC	Fe, Ni, Co, Mn	—
ZrB ₂ , TiB ₂	Ni, Cr	—
ThO ₂	F	Ca
ZrO ₂	—	H ₂ , Cr, Ti, Ni, Mn
BaTiO ₃	—	Ti, Ta, Al/Si/Ti
Y ₂ O ₃	—	Th
Pb(ZrTi)O ₃	—	Al, Fe, Ta, La
ZnFe ₂ O ₃	BiO ₃ , SiO ₂ , CaO	—

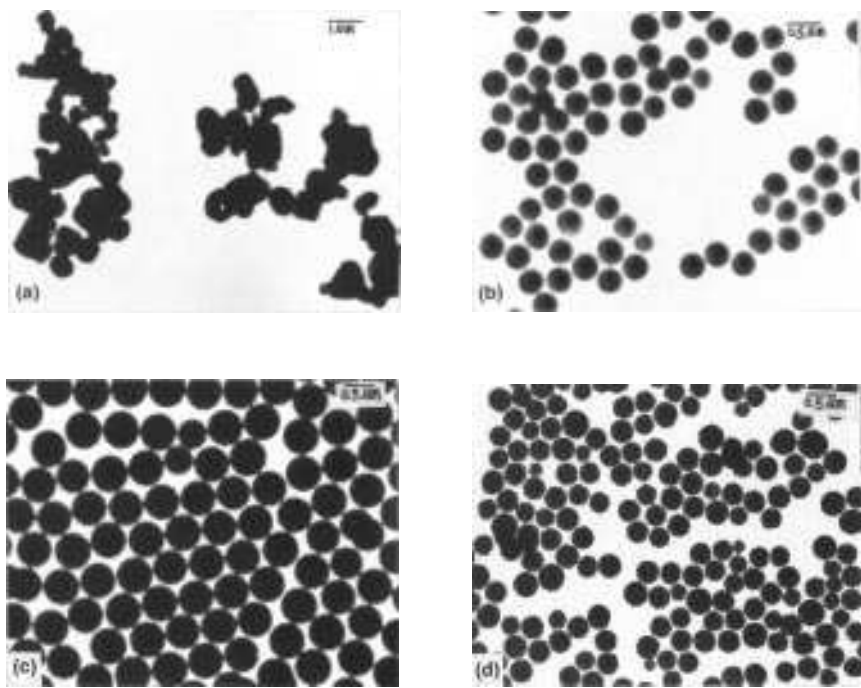


Figure 3. TEM micrographs of particles with various coatings. (a) Commercial TiO₂ coated with 3 wt% of SiO₂; (b) Y₂O₃ coated with 50 wt% of SiO₂; (c) SiO₂ with 10 wt% of Y₂O₃; (d) TiO₂ coated with 5 wt% of SiO₂.

4. Powder II: starting powders manufactured by nonconventional ceramic methods

Besides the conventional ceramic methods, there are a variety of other physical and wet-chemical methods to synthesize fine particle methods reported in literature.

Unlike the conventional ceramic methods, other physical methods are basically buildup methods [55]. This method involves the generation of particles by agglomeration (buildup) of atoms or molecules of the individual elements concerned. The method itself depends on the heating and evaporation of selected raw materials to reduce them to their atomic or molecular form. This is followed by multiple condensation to generate the required particles. Generally, the size of particles is in the order of 10–100 nm. The number of atoms or molecules required to generate a single particle is in the region of 4×10^3 for the lower limit of 5 nm, and somewhere around 3×10^7 for the particle with the size of 100 nm.

Over a number of years researchers have used not only gas evaporation but also a variety of new methods and mechanisms to generate fine particles, which include rf and dc Sputtering [56] and Plasma Deposition [57] etc.

Siegel [3–5] and others [58] have prepared nanophase ceramics by the gas-condensation method. Nanophase ceramic powders possess distinct advantages including (i) enhanced sinterability at lower temperature resulting from high driving forces and short diffusion distances, (ii) high surface and grain boundary purity due to the relative cleanliness of the gas-condensation technique, and (iii) when consolidated, potential superior properties from corresponding micro-phase materials, owing to the fact that a large fraction of ions resides in interfaces [3]. The ceramic particles obtained in their experiments, for example, ZnO with an average diameter of 13 nm, can be sintered at 600°C.

Chemical methods for powder preparation in high quality ceramics become very popular, especially in the electronic materials industry. Despite their initially higher cost than conventional solid-state reaction of mechanically mixed and calcined starting materials, the improved performance and reproducibility achievable with chemically derived powders ultimately saves money and adds value. Many published papers and patents are about the synthesis of fine ceramic powders by chemical methods. Usually, the wet-chemical methods can be divided into the following categories: coprecipitation [59–61], sol-gel [62], controlled double-jet precipitation [63–66], hydrothermal process [67–73], and homogeneous precipitation [53] etc.

Coprecipitation and hydrothermal processes are widely used in producing fine particles. In electronic materials industry, Ni-Zn ferrite particles synthesized by coprecipitation have the average size about 100 nm and the lower corresponding sintering temperature was also reported [59–61]. The advantages of chemical coprecipitation include technically simple, low manufacturing cost, high reproducibility and fine particle size; the disadvantages are difficult control of final chemical composition of products, agglomeration of particles, design of reactor and scaling-up.

As far as hydrothermal process is concerned, it is quite promising. The hydrothermal process is a typical buildup method, where fine particles are made from much smaller units such as clusters, molecules, ions and atoms. Moreover, the particle products of hydrothermal process are normally crystalline and do not need presintering for conversion from amorphous to crystalline as needed in most other wet-chemical process. With the over-increasing energy cost, the hydrothermal method becomes attractive for fine powder preparation because of

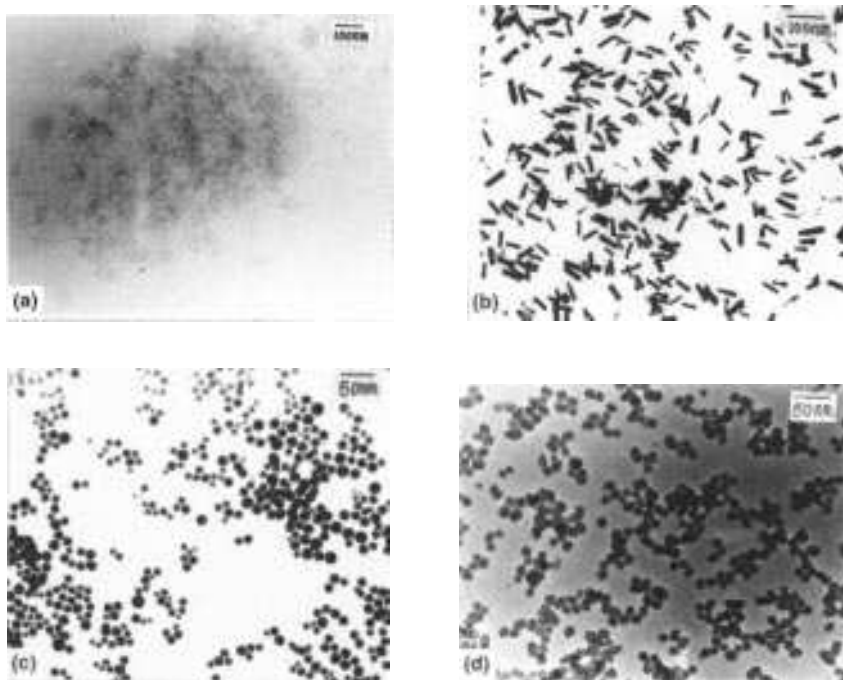


Figure 4. Microphotographs of some typical particle products by hydrothermal process. (a) γ -AlOOH; (b) γ -FeOOH; (c) BaTiO₃; (d) ZrO₂.

the low temperature involved and the good sinterability [68]. The normal batch process in the industrial pilot scale can produce 1 kg powders using 10 L autoclaves within 3 hours (30 minutes for the preheating, 2 hours of reaction and 30 minutes for post treatment). Microphotographs of some typical particle products by hydrothermal process are shown in figure 4. Ultrafine BaTiO₃ powders produced by hydrothermal process are sintered to 95% of theoretical density at 1150°C for 1 hour.

Figure 5 is the fractured surface of sintered BaTiO₃. After sintering, the BaTiO₃ compacted powders densified with an average grain size of 1.7 μ m, and pores closed to form well-defined spherical micropores with a diameter of 0.5 μ m.

In addition, the hydrothermal reaction can be designed as a continuous process by tubular reactor. The schematic flow diagram of continuous hydrothermal process is illustrated in figure 6. In the continuous hydrothermal process, the hydrothermal precursors are preheated to the desired temperature, and pumped into a titanium alloy tubing reactor to carry out the reaction. In the other end, particle suspensions are cooled, washed and separated. The temperature is normally around 200°C and the water vapor pressure is about 15 atm. The successful example of continuous hydrothermal process can produce Al₂O₃ and ZrO₂ etc. ceramic powders with an average diameter <100 nm.

Recently, Controlled Double-Jet Precipitation (CDJP) was successfully introduced to make uniform nanosize particles. Her and co-workers [63] reported producing uniform spherical BaTiO₃ particles with a mean diameter about 220 nm. However, unlike similar

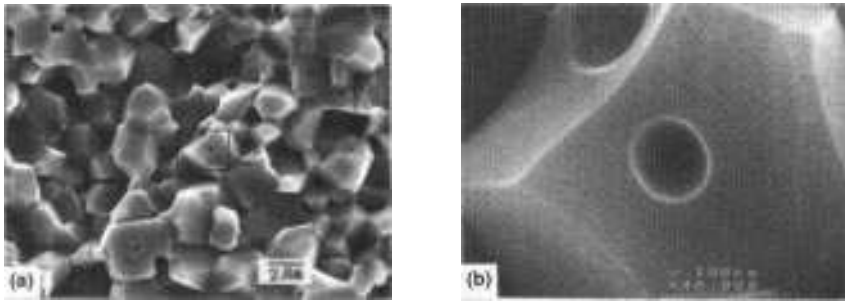


Figure 5. Fracture surface of sintered BaTiO₃ with well-defined closed spherical micropores, where (b) is the enlargement of a micropore in center of picture (a).

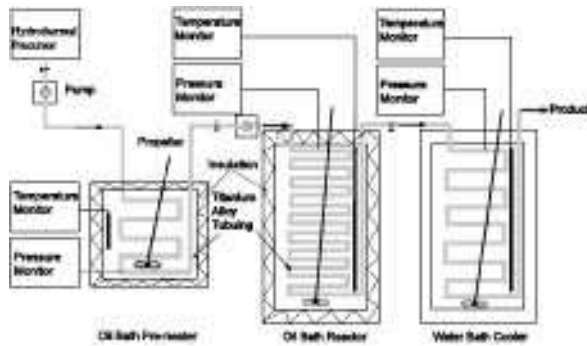


Figure 6. Schematic flow diagram of continuous hydrothermal process.

particles produced by other methods, the BaTiO₃ particles produced by CDJP techniques in their paper are apparently the agglomerates or aggregates of nanosize subunits with average size about 5–10 nm. The special microstructure is determined by the unique particle formation mechanisms associated with CDJP technique [64]. In their paper, BaTiO₃ was sintered to almost theoretical density at 1250°C, while the commercial BaTiO₃ particles reached only 90% of the theoretical density (figure 7).

5. Pressed compaction

Powders are consolidated into approximately the desired shape by a variety of methods, one of the most common being pressed in a die. Pressures from about 10,000 psi (~70 MPa) to 100,000 psi (~700 MPa) are commonly used. Too high pressing pressures leave residual stresses in the compact that can lead to cracking before sintering. Organic binders are used in many pressing operations to lubricate the powder particles so an uniform density is attained during pressing, as well as to give strength to the green body.

Higher green body density will improve the sinterability of ferrites. A green body with higher density increases the contact area of neighboring particles and reduces the volume of micro-pores, promotes the densification process and reduces the sintering temperature.

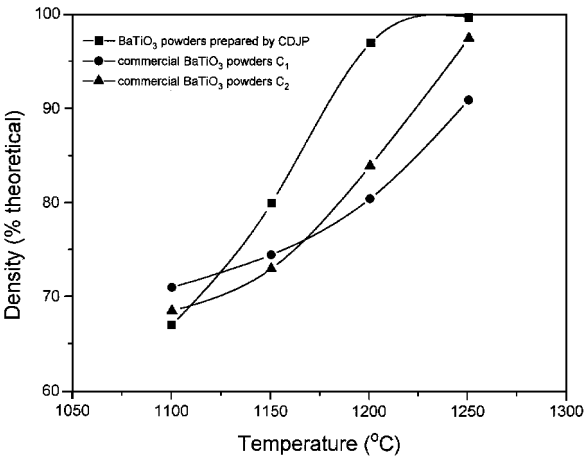


Figure 7. Plot of the density as a function of sintering temperature of the BaTiO₃ powders. ■: Powders prepared by CDJP; ▲: commercial powders C₁; and ●: commercial powders C₂.

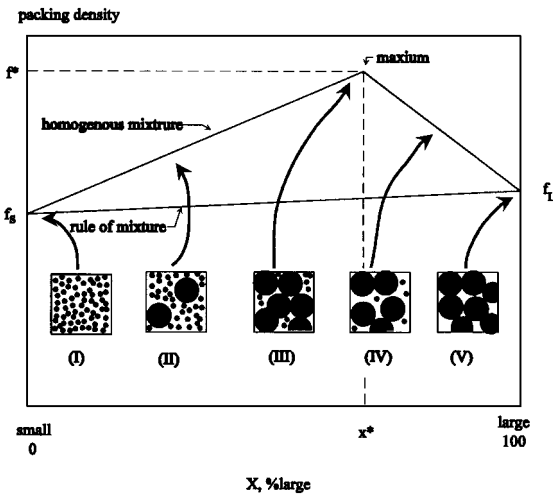


Figure 8. Packing density of a binary particle sizes system.

The high green body density can be obtained by combining moderate pressing pressure with the right use of binder, and the right combination of particles with different sizes (figure 8). Besides using higher pressing pressure in the modeling process, there are two new techniques reported in literature. The first one employed ultrasound during the compaction process and observed improved behavior of compaction when applied pressure is under critical value [74]. The other one used wet-pressing to avoid the agglomeration formed during the spray-drying process after ball-milling and improved the sinterability [75].

Furthermore, the gradients in the packing density of powder in a compact can cause stresses to develop as sintering proceeds; the consequence may be locally incomplete

sintering, warping of the piece, or internal cracks. The classic situation is seen in a cylindrical piece pressed in a die. Because of friction between powder and the die wall, the packing density in the center of the piece is lower than that near the die punches; this leads to an hourglass shape in the final piece. If the outer shell sinters sufficiently at first, it may become strong enough to prevent this deformation shrinkage; fissures then develop in the center of the specimen as sintering proceeds.

6. Sintering

Although it is possible to reduce the sintering temperature by prolonging sintering time, the consequent risk of over firing, excessive grain growth and possible higher production cost prohibits the use of longer sintering time.

Hot-pressing or sintering under pressure can reduce the sintering temperature [2, 26, 28]. In this case, the driving force for sintering is increased and higher density can usually be obtained at a low sintering temperature. The increased driving force arises from the local increase in the free energy of the matter in the highly stressed regions where particles make contact. It has been established that the energy available for densification is increased by more than 20% by the application of pressure during sintering. The induced several processing and property advantages include: 1) reduces densification time; 2) reduces densification temperature, often resulting in less grain growth than would occur with pressureless sintering; 3) minimizes residual porosity, and 4) results higher mechanical strength than that can be achieved through pressureless sintering, due to the minimization of porosity and grain growth [2]. Hot pressing is particularly advantageous if a fine grain size is desired in a sintered ceramic, because less grain growth occurs at lower sintering temperature. It is known in practice that under a pressure of 150–200 atm and at a temperature of approximately 1800°C, pure alumina can be sintered to a practically completely nonporous body (having theoretical density) within minutes. Whereas without pressure application appreciably higher temperature and hours-long sintering time is required to obtain a product still having 1–2 vol% porosity. Moreover, a hot-pressed alumina body has an almost homogeneous microstructure, in some respects resembling that of a monocrystalline sapphire or ruby.

Table 3 lists a comparison of densities and strengths achieved by hot-pressing versus sintering.

In recent years, the hot isostatic process (HIP) was also used. In it, the ceramic is commonly sintered to the closed-pore stage and then placed in a hot chamber and subjected to helium gas pressure of several thousand psi.

Both these processes are very expensive and are little employed in industrial scale. Cemented tungsten carbide and jet engine components made from rapidly solidified superalloy powder are valuable enough products that HIP is economically feasible.

7. Optimization of sintering process

As stated before, the ultimate goal of research is to get the highest quality of a product at lowest cost. It is highly desirable if the whole sintering process can be optimized [75, 76]. Gropyanov and Abbakymov [75] employed non-isothermal kinetics methods in

Table 3. Comparison of densities and strengths achieved by hot pressing versus sintering.

Material	Sintering aids	Density (% theoretical)	RT MOR ^a (Mpa)	1350°C MOR (Mpa)
Hot-pressed Si ₃ N ₄	5%MgO	>98	587	173
Sintered Si ₃ N ₄	5%MgO	~90	483	138
Hot-pressed Si ₃ N ₄	1%MgO	>99	952	414
Sintered Si ₃ N ₄	BeSiN ₂ + SiO ₂	>99	560	—
Hot-pressed Si ₃ N ₄	13%Y ₂ O ₃	>99	897	669
Sintered Si ₃ N ₄	6%Y ₂ O ₃	~98	587	414

^aRT MOR—Room temperature modulus of rupture.

the description of the densification and grain-size growth processes during sintering of Al₂O₃, and obtained a mathematical model of these processes that fitted the whole experimental data. It generated the idea to optimize the ceramic sintering process with the aim of achieving predetermined parameters of its structure (mean grain size—*l* or porosity—*p*) and consequently its other physical properties. Their unique sintering description allows them to correlate densification rate or grain size growth rate with some empirical constants without considering the sintering mechanism [77]. Thus, kinetics equation adequacy for experimental data is of primary importance [78].

Gropyanov and Rizhikov [79] published a model of ceramics abrasive wear that connects the energy parameter of wear resistance with mean grain size and relative volume of pore reduction (*Z*):

$$I_w^{-1} = K_w l^{-a} e^{-bZ} \tag{4}$$

and,

$$Z = \frac{p}{(1 - P)} \frac{(1 - P_0)}{P_0} \tag{5}$$

where I_w^{-1} is energy wear resistance, K_w and b are constants, Z is the parameter of densification during sintering, P_0 and P are the initial porosity and porosity after sintering. There exists an exponential dependence of wear from porosity in [80]:

$$I = e^{-bP} \tag{6}$$

where I —wear volume, P —porosity and b —constant for a given type of porosity. The introduction of parameter Z in above equations makes it possible to describe the wear resistance dependence on porosity for a wide region of porosity values [81]. The mathematical model of ceramics wear dependence on structure-sensitive properties formed during the sintering process makes it possible to compute the optimal temperature-time conditions of sintering, in regard to achieving maximum resistance of abrasive wear. They used 99% α -Al₂O₃ with the addition of MgO as modifiers in their experiments for comparison. The resulted experimental data fitted with the predicted parameters from the optimized model.

The problem associated with the optimization process is how to quantify the properties of sintered body with the process parameters, before any mathematical work can be done. Especially in the case of electronic materials, because the relationship between electronic

properties and sintering conditions is too complicated to put in simple mathematical expressions. Moreover, the effects from minor impurities are very difficult to control. However, there is another practical possibility to reach an optimal sintering condition to achieve the proposed properties. For example, the property in question is Q , and Q is a function of sintering conditions as expressed in following:

$$Q = f(a, b, c, d, e, f) \quad (7)$$

Where a, b, c, d, e, f are particle size, size distribution, sintering temperature, green body density, additives and surface energy respectively. Starting from $Q_{a0, b0, c0, d0, e0, f0}^0$, only one parameter (for instance, a in this case) is changed at one time and reach a local optimum $Q_{a1, b0, c0, d0, e0, f0}^1$, and then change the second parameter (for example, b in this case), and locate another local optimum point $Q_{a1, b1, c0, d0, e0, f0}^2$, and repeat four more times to reach $Q_{a1, b1, c1, d1, e1, f1}^6$. The first set of searches is finished. One can conduct the second set search by doing the same. Mathematically speaking, above principle can be expressed as:

First search:

$$\left. \frac{\partial Q}{\partial a} \right|_{b_0, c_0, d_0, e_0, f_0} = 0 \quad (8)$$

and based on the same principle, the next five searches can be performed at the similar way, and finally the local optimum Q_1 is reached at the sintering conditions of a_1, b_1, c_1, d_1, e_1 , and f_1 . After repeat above steps and conducted n times of searches or n sets of experiments, when the $Q_n \cong Q_{n+1}$, and the final optimal sintering conditions are $a_{n+1}, b_{n+1}, c_{n+1}, d_{n+1}, e_{n+1}$, and f_{n+1} . However, in practice is not likely to use this method while considering all possible sintering variables, because of a huge amount of work and also possible divergence. Combining with experience to eliminate some variables, this method may make significant contribution to lower sintering temperature and reduce the production cost while keep a high quality.

8. Applications in low temperature co-fired ceramics (LTCC)

In the past 20 years, Low Temperature Co-fired Ceramics were developed and improved in semiconductor and micro-electronic industry. The main reason was that, they had to use hybrid circuit and required to fire ceramics at less than 1000°C if gold was used as the conductor, and less than 900°C if silver was used as the conductor line.

There are many successful application examples as listed in Refs. [82–90]. Two of them are discussed as follows.

First, the low temperature sintering of NiZn ferrite was a good example. NiZn ferrite is one of the most important materials widely used in electronic components. The normal sintering temperature of NiZn ferrite is around 1000°C to 1250°C. However, with the development of surface mounting hybrid components and chip inductors, NiZn ferrite are required to fired at the temperature less than the melting point of silver (962°C). The use of CuO as an additive successfully reduced the sintering temperature from 1000°C and higher to around 900°C. Later, Bi₂O₃ and lead glass were introduced to reduce the sintering temperature even further to 875°C [82–87].

Second, the low temperature sintering of BaTiO_3 was another good example. BaTiO_3 was doped with Ca, Sr and Zr to achieve the lower sintering temperature. The composition of the doped BaTiO_3 can be expressed as $(\text{Ba}_{1-x-y}\text{Ca}_x\text{Sr}_y)\text{O}_m(\text{Ti}_{1-z}\text{Zr}_z)\text{O}_2$ [88], wherein $0 \leq x \leq 0.25$, $0 \leq y \leq 0.05$, $0.1 \leq z \leq 0.3$, and $1.000 \leq m \leq 1.020$. The resulted dielectric ceramic can be co-fired with Ag/Pd thick film printed electrode.

Interested reader may refer to Refs. [82–90] for more information in this topic.

9. Conclusion

In order to improve the sinterability while maintain the high product quality standard, the followings are suggested:

- a: From the starting commercial powders obtained by conventional ceramic methods, the powders can be pretreated either by surface activation or introducing second phase in the form of coating or additives, and then powders are hot-pressed to meet specific requirements. Furthermore, the ultrafine grinding and grinding aids can improve the grindability, reduce the ground particle size, and prevent the reagglomeration of particles so that the sintering temperature can be feasibly reduced.
- b: From the starting ceramic powders synthesized from wet-chemical method, coprecipitation and hydrothermal processes are suggested for the production of nano-size and agglomeration-free ceramic precursor particles; while Controlled Double-Jet Precipitation (CDJP) may be employed in some cases. If possible, the powders can be pretreated by surface activation or coating before conventional sintering or hot-pressing.

However, it has to be reminded that, although wet-chemical powder methods are possible to produce nano-size and agglomeration-free particles, the initial single investment and process maintenance fee are much higher than that of the conventional ceramic method. This may help explain why the conventional ceramic method is dominating the electronic ceramic industry right now. Out of the purpose of faster development and production at lower cost, in another word, more profit and improved competitiveness, the selection to improve the sinterability or reduce sintering temperature must be based on the existing equipment and technology, and should be product-oriented.

References

1. Upadhyaya, G.S.: *Science of Sintering*, 27(2):49–70, 1995.
2. Rosolowski, J.H. and Burke, J.E.: In *Encyclopedia of Chemical Processing and Design*, John J. McKetta (ed.), Chapter of Sintering, New York: Marcel Dekker, Inc., pp. 301–329, 1995.
3. Lee, J., Hwang, J.-H., Mashek, J.J., Mason, T.O., Miller, A.E., and Siegel, R.W.: *J. Mater. Res.*, 10(9):2295–2300, 1995.
4. Siegel, R.W., Ramasamy, S., Hahn, H., Zongquan Li, Ting Lu, and Gronski, R.: *J. Mater. Res.*, 3:1367, 1988.
5. Siegel, R.W.: *Mater. Sci. Eng.*, A 168:189, 1993.
6. Stuijts, A.L.: *Sintering and related phenomena. Proc. of 3rd Int. Conference*, G.C. Kuczynski (ed.), New York: Plenum Press, pp. 331–337, 1973.
7. Hausner, H.H.: In *Sintering New Development*, M.M. Ristic (ed.), Amsterdam: Elsevier Scientific Publishing Co., p. 3, 1979.

8. Kuczynski, G.C.: In *Sintering New Development*. M.M. Ristic (ed.), Amsterdam: Elsevier Scientific Publishing Co., p. 245, 1979.
9. Ristic, M.M.: *Int. Symp. on Science and Technology of Sintering "Sintering 87,"* Tokyo, 1987.
10. Ristic, M.M.: *Science of Sintering*, 22(1):3–9, 1990.
11. Lenel, F.V. and Ansell, G.S.: *Modern Development in Powders Metallurgy*, H.H. Hausner (ed.), New York: Plenum Press, Vol. 1, p. 281, 1966.
12. Lenel, F.V., Ansell, G.S., and Morris, R.C.: *Modern Development in Powders Metallurgy*, H.H. Hausner (ed.), New York: Plenum Press, Vol. 4, p. 199, 1971.
13. Schatt, W. and Freidrich, E.: *Crystal Research and Technology*, 17(9):1061, 1982.
14. Schatt, W., Freidrich, E., and Wieters, K.P.: *Reviews of Powder Metallurgy and Physical Ceramics*, 3:1–111, 1986.
15. Kuczynski, G.C.: *J. Appl. Physics*, 20:1160, 1949.
16. Kuczynski, G.C.: *J. Appl. Physics*, 20:632, 1950.
17. Gouri, S., Lund, J.A., and Chaklader, A.C.D.: *Solid State Phenomena*, 25/26:45–54, 1992.
18. Chu, M.-Y., Rahaman, N.N., Jonghe, L.C.De, and Brook, R.J.: *J. Am. Ceram. Soc.*, 74:1217–1225, 1991.
19. Hassold, G.N., Chen, I.-W., and Srolovitz, D.J.: *J. Am. Ceram. Soc.*, 73:2857–2864, 1990.
20. Chen, I.-W., Hassold, G.N., and Srolovitz, D.J.: *J. Am. Ceram. Soc.*, 73:2865–2872, 1990.
21. Madhavrao, L.R. and Rajagopalan, R.: *J. Mater. Res.*, 4(5):1251–1256, 1989.
22. Ristic, M.M., Nikolic, M.V., and Radic, S.M.: *Science of Sintering*, 26(2):89–106, 1994.
23. Samsonov, G.V., Pryadko, I.F., and Pryadko, L.F.: *A Configuration Model of Matter*, New York: Consultants Bureau, 1973.
24. German, R.M.: *Science of Sintering*, 15:27, 1983.
25. Uphadhyaya, G.: *Int. Symp. on Science of Sintering "Sintering 87,"* Tokyo, 1987.
26. Richerson, D.W.: *Modern Ceramic Engineering: Properties, Processing and Use in Design*, New York: Marcel Dekker, p. 216, 1988.
27. Snelling, E.C.: *Soft Ferrites: Properties and Applications*, London: Iliffe Books Ltd., pp. 1–17, 1988.
28. Ryshkewitch, E.: *Oxide Ceramics, Physical Chemistry and Technology*, New York: Academic Press, pp. 44–67, 1960.
29. Fuerstenau, D.W.: *Kona (Powder)*, 13:5–17, 1995.
30. Lowrison, G.C.: *Crushing and Grinding*, Cleveland, OH: CRC Press, Inc., p. 60, 1979.
31. El-Shall, H. and Somasundaran, P.: *Powder Technology*, 38:275–293, 1984.
32. Somasundaran, P. and El-Shall, H.: In *Ultrafine Grinding and Separation of Industrial Minerals*, S.G. Malghan (ed.), New York: AIME, pp. 21–35, 1983.
33. El-Shall, H. and Somasundaran, P.: *Powder Technology*, 38:267–273, 1984.
34. Schoenert, K.: *Trans. Soc. Min. Engineers, AIME*, 252:21–26, 1972.
35. Hammond, M.L. and Ravitz, S.F.: *J. Am. Ceram. Soc.*, 46:329–332, 1963.
36. Velamaka, B.V. and Fuerstenau, D.W.: *Floculation in Biotechnology and Separation System*, Y.A. Attia (ed.), Amsterdam: Elsevier, pp. 211–223, 1987.
37. Fuerstenau, D.W., Venkataraman, K.S., and Velamakanni, B.V.: *Int. J. Mineral Processing*, 15:251–267, 1985.
38. Fuerstenau, D.W., Kapur, P.C., and Velamakanni, B.V.: *Int. J. Mineral Processing*, 28:81–98, 1990.
39. Kapur, P.C., Ranjan, S., and Fuerstenau, D.W.: *Int. J. Mineral Processing*, pp. 9–29, 1992.
40. Klimpel, R.R.: *Powder Technology*, 31:255–262, 1982.
41. Klimpel, R.R.: *Powder Technology*, 32:267–277, 1982.
42. Gao, M.W. and Forsberg, E.: *XVIII Int. Process. Congr.*, Sydney, Australia, pp. 237–244, 1993.
43. Laskowski, J.S. and Pugh, R.J.: *Colloid Chemistry in Mineral Processing*, Elsevier, Chapter 4, pp. 115–171, 1992.
44. Bucall, R. and Corner, T.: *Colloids Surf.*, 17:39–44, 1986.
45. Vincent, B. and Whittington, S.G.: *Surface and Colloid Science*, E. Matijevic (ed.), Plenum Press, Vol. 12, pp. 1–108, 1982.
46. Tanaka, T.: *Kona (Powder)*, 13:19–28, 1995.
47. Kostic, E., Kis, S., and Boskovic, S.: *Science of Sintering*, 26(1):11–20, 1994.

48. Buchanan, R.C.: *Ceramic Materials for Electronics: Processing, properties and applications*, New York: Marcel Dekker, 1986.
49. Fang, T.T., Hwang, J.B., and Shiau, F.S.: *J. Mater. Sci. Lett.*, 8:1386–1388, 1989.
50. Giesche, H. and Matijević, E.: *J. Mater. Res.*, 9(2):436–450, 1994.
51. Mitra, R., Puri, R.K., and Mendiratta, R.G.: *J. Mater. Sci.*, 27:1275–1279, 1992.
52. Mitra, R., Puri, R.K., and Mendiratta, R.G.: *J. Mater. Sci. Lett.*, 11:276–277, 1992.
53. Matijević, E.: *Chem. Mater.*, 5:412–426, 1993.
54. Risbud, S.H. et al.: *J. Mater. Res.*, 10(2):237–239, 1995.
55. Ichinose, N., Ozaki, Y., and Kashu, S.: *Superfine Particle Technology*, New York: Springer-Verlag, p. 80, 1988.
56. Williams, C.M. et al.: *J. Appl. Phys.*, 75(2):1676–1680, 1994.
57. De Marco, M.: *J. Appl. Phys.*, 73(10):6287–6289, 1993.
58. Hahn, H., Logas, J., and Averback, R.S.: *J. Mater. Res.*, 5:609, 1990.
59. Kodama, T. et al.: *J. Appl. Phys.*, 69(8):5915–5917, 1991.
60. Chang, C.W., Tzeng, M.S., and Wang, S.J.: *J. Mater. Sci. Lett.*, (9):832–835, 1990.
61. Pandya, P.B., Joshi, H.H., and Kulkarni, R.G.: *J. Mater. Sci.*, 26:5509–5512, 1991.
62. Rozman, M. and Drofenik, M.: *J. Am. Ceram. Soc.*, 28(9):2449–2455, 1995.
63. Her, Y.-S., Matijević, E., and Chon, M.C.: *J. Mater. Res.*, 10(12):3106–3114, 1995.
64. Stavek, J., Sipek, M., Hirasawa, I., and Toyokura, K.: *Chem. Mater.*, 4:545–555, 1992.
65. Sugimoto, T.: *Adv. Colloid Interface Sci.*, 28:65–108, 1987.
66. Wey, J.S.: *Preparation and Properties of Solid State Materials*, New York: Marcer Dekker, vol. 6, pp. 67–117, 1981.
67. Komarneni, S., Fregeau, E., Breval, E., and Roy, R.: *J. Am. Ceram. Soc.*, 71(1):C26–28, 1988.
68. Pannaparayil, T., Marande, R., and Komarneni, S.: *J. Appl. Phys.*, 69(8):5349–5351, 1991.
69. Pannaparayil, T., Marande, R., Komarneni, S., and Zadarko, M.: *J. Appl. Phys.*, 67(9):5509–5511, 1990.
70. Dawson, W.J.: *Am. Ceram. Soc. Bull.*, 67(19):1673–1678, 1988.
71. Kuang, X. and Mao, M.: *Engineering Chemistry and Metallurgy (Chinese)*, 4:85, 1994.
72. Kuang, X., Carotenuto, G., Nicholais, L. et al.: *Journal of Applied Composite Materials*, 1:1–11, 1996.
73. Kuang, X., Mao, M. et al.: *Proceeding of the 5th International Symposium on Ceramic Materials & Components for Engines*, Singapore: World Scientific Publishing Ltd., 1995.
74. Emeruwa, E., Jarrige, J., and Mexmain, J.: *J. Euro-Ceram.*, 1(1):248–252, 1989.
75. Hsieh, H.L. and Fang, T.T.: *J. Am. Ceram. Soc.*, 72:142, 1989.
76. Schatt, W.: *Science of Sintering*, 25(3):99–107, 1993.
77. Gropyanov, V.M. and Rizhikov, A.L.: *Science of Sintering*, 25(3):109–116, 1993.
78. Ivensen, V.A.: *Metallurgiya*, 247, 1985.
79. Gropyanov, V.M. and Abbakymov, V.G.: *Poroshkaya Metallurgiya*, 5:76, 1975.
80. Wu, C.C. and Rice, R.W.: *Ceram. Eng. and Sci. Proc.*, 6:977, 1985.
81. Rizhikov, A.L. and Gropyanov, V.M.: *Soviet J. of Friction and Wear*, 2:361, 1991.
82. Hsu, J.Y., Ko, W.S., Shen, H.D., and Chen, C.J.: *IEEE Transaction on Magnetics*, 30(6):4875–4877, 1994.
83. Takaya, M.: *Japanese Electronic Engineer.*, pp. 38–41, 1986.
84. Nomura, T. and Nakano, A.: *Proceedings of the 6th International Conference on Ferrites*, T. Yamaguchi and M. Abe (eds.), Tokyo and Kyoto, Japan: The Japanese Society of Powder and Powder Metallurgy, pp. 1198–1201, 1992.
85. Nakano, A., Momoi, H., and Nomura, T.: *Proceedings of the 6th International Conference on Ferrites*, T. Yamaguchi and M. Abe (eds.), Tokyo and Kyoto, Japan: The Japanese Society of Powder and Powder Metallurgy, pp. 1225–1228, 1992.
86. Goldman, A.: *Chemical Aspects of Ferrites*, in *Modern Ferrite Technology*, New York: Van Nostrand Reinhold, Chap. 3, pp. 87–114, 1990.
87. Tormey, E.S.: *IEPS. Proceedings of the Technical Conference, International Electronics Packaging Conference*, pp. 215–228, 1990.
88. Taguchi, T., Kuribayashi, H., Tanaka, J., Konage, T., and Morigami, Y.: *US Patent No. 05468694* please also see *US Patent No. 4396720*, *4519828*, *4528275*, *4654095*, *4655864*, *4849379*, *4849380* and *5356841*.
89. Nomura, T., Nakano, Y., Satoh, A., and Hitomi, A.: *US Patent No. 05319517*.
90. Prabhu, A.N., Conlon, E.J., and Thaler, B.J.: *US Patent 05216207*. Please also see *US Patent No. 4355115*, *4733018*, *4748136*, *4772574*, *4808673*, *48410420*, *4816615*, *4874550*, *4880567* and *5024883*.